

KCPC Week 2 Problems

Heuristics, Advanced Topics, and Algorithmic Bridges

How To Use This Sheet

Start with the heuristic problems to complete the Pair 1 curriculum. The advanced and algorithmic sections then deepen those ideas if time permits. For algorithmic bridge problems, sketch the algorithm and argue correctness before writing code.

1 Wishful Thinking

Basic – Complete The Square

For all real numbers x and y , prove that

$$x^2 + y^2 \geq \frac{(x + y)^2}{2}.$$

Try to rewrite the difference as something that is clearly non-negative.

Easy – Make The Sum Telescope

Prove that

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}.$$

Wish that each term were a difference of two simpler fractions.

Medium – A Number Puzzle

Construct infinitely many integer solutions to $x^2 + y^2 = 2z^2$ with $x, y, z > 0$. Wish that a Pythagorean triple were hidden in the equation.

Hard – Improve The Recurrence

Let $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Find a closed formula for a_n . Wish that the unwanted $+1$ were absorbed into the sequence.

Very Hard – Factor The Equation

For a fixed positive integer n , find all positive integer pairs (x, y) satisfying

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}.$$

Wish that the equation could be written as a product.

2 Symmetry

Basic – Chomp: Poisoned Chocolate

On an $m \times n$ rectangular chocolate bar, the bottom-left square is poisoned. Players alternately choose a square and eat it together with all squares above and to its right. The player forced to eat the poisoned square loses. Show that the first player has a winning strategy for any $m, n \geq 2$. Use a strategy-stealing argument.

Easy – Coins On A Circular Table

Two players alternately place identical non-overlapping circular coins on a circular table. The player who cannot move loses. Show that the first player has a winning strategy.

Medium – A Symmetric Integral

An integrable function $f : [0, 1] \rightarrow \mathbb{R}$ satisfies $f(x) + f(1 - x) = 1$ for every $x \in [0, 1]$. Evaluate $\int_0^1 f(x) dx$.

Hard – Odd Skew-Symmetric Matrices

Let A be an $n \times n$ real matrix satisfying $A^T = -A$. Prove that if n is odd, then $\det A = 0$.

Very Hard – Six People

Among any six people, prove that there are either three mutual acquaintances or three mutual strangers.

3 Extreme Principle

Basic – Erdős–Szekeres Monotone Subsequence

Prove that any sequence of $n^2 + 1$ distinct real numbers contains an increasing or decreasing subsequence of length $n + 1$. (Hint: associate each term with the length of the longest increasing subsequence starting there.)

Easy – Minimum Degree Forces A Cycle

Prove that every finite graph in which every vertex has degree at least 2 contains a cycle. Start with a longest path.

Medium – An Empty Triangle

A finite set of points in the plane is not contained in one line. Prove that three of the points form a triangle whose interior contains none of the other points.

Hard – A Path Through A Tournament

In a tournament, every pair of vertices is joined by exactly one directed edge. Prove that every finite tournament has a directed path that visits every vertex exactly once.

Very Hard – Longest Paths Meet

Prove that any two longest paths in a finite connected graph share at least one vertex.

4 Reduction

Basic – An Acquaintance Graph

At a party of $n \geq 2$ people, acquaintance is mutual. Prove that two people have the same number of acquaintances at the party. Reduce the problem to a graph and consider the possible vertex degrees.

Easy – The Bridges Of Königsberg

Reduce the bridges puzzle to a graph and prove that no walk crosses each of the seven bridges exactly once.

Medium – Missionaries And Cannibals

Three missionaries and three cannibals must cross a river using a boat that carries at most two people. On a bank with at least one missionary, cannibals must never outnumber missionaries. Model the problem as a state graph and find a safe crossing sequence.

Hard – Lamps As Linear Algebra

n lamps are arranged in a circle. A move toggles lamp i and its two neighbours. Given an initial configuration, give an algorithm to decide whether all lamps can be turned off.

Very Hard – Assignment As Maximum Flow

There are n students and n projects. Each student supplies a list of acceptable projects. Build a flow network for assigning every student a distinct acceptable project. State how its maximum flow determines whether an assignment exists.

Advanced Material

The remaining problems are optional extensions. Their theory and worked examples are presented in *Week 2 Advanced Slides*.

5 Advanced Logic And Common Knowledge

Basic – Blue-Eyed Islanders ($k = 2$)

There are 2 islanders with blue eyes, and the rest have brown. They cannot see their own eyes. A trusted outsider announces: “At least one of you has blue eyes.” Each day, anyone who knows their own eye colour leaves at midnight. What happens and on which day?

Easy – Cheating Husbands

In a village of 100 couples, some husbands have been unfaithful. Every wife knows which *other* husbands have cheated, but not her own. The queen announces: “At least one husband has been unfaithful.” If a wife deduces her husband cheated, she must act at midnight. What happens?

Medium – Sum And Product Puzzle

Two integers x, y with $2 \leq x < y$ and $x + y \leq 100$. Mr P knows the product xy ; Mr S knows the sum $x + y$. P: “I don’t know the numbers.” S: “I knew you didn’t know.” P: “Now I know the numbers.” S: “Now I know them too.” Find x and y .

Hard – Cheryl’s Birthday

Cheryl gives Albert and Bernard this list of 10 possible dates: May 15, May 16, May 19; June 17, June 18; July 14, July 16; August 14, August 15, August 17. She whispers the month to Albert and the day to Bernard. Albert: “I don’t know the birthday, but I know Bernard doesn’t know either.” Bernard: “At first I didn’t know, but now I do.” Albert: “Now I also know.” When is Cheryl’s birthday?

Very Hard – 100 Prisoners And A Light Bulb

100 prisoners are in solitary cells. There is a central room with a light bulb (initially off) and a switch. One prisoner at a time is chosen independently and uniformly at random to visit the room; they may toggle the switch and/or announce “everyone has visited.” If correct, all go free; if wrong, all die. They may agree a strategy beforehand. Design a protocol that never announces incorrectly and eventually frees everyone with probability 1.

6 Advanced Invariants And Monovariants

Basic – Peg Solitaire (Small Board)

On a 3×3 board with the centre empty, pegs jump orthogonally over adjacent pegs into empty squares, removing the jumped peg. Can you finish with one peg in the centre?

Easy – Token Game

Tokens on integer points. A move: choose x with tokens at x and $x + 1$; remove them and place one at $x + 2$. Start with one token at 0 and one at 1. Can you ever have a token at 100? How many moves are possible?

Medium – Conway's Soldiers

On an infinite checkerboard with $x, y \in \mathbb{Z}$, pegs occupy all squares with $y \leq 0$. A peg jumps orthogonally over another into an empty square, removing the jumped peg. Can a peg ever reach $y = 5$? (Hint: for a target $(x_0, 5)$, use weights $\varphi^{|x-x_0|+|y-5|}$ where $\varphi = (\sqrt{5} - 1)/2$.)

Hard – Polynomial Invariant For (x+2,y-1)

A token starts at $(0, 0)$ on the integer grid $\mathbb{Z}^2$. A move changes (x, y) to $(x + 2, y - 1)$. Find a non-constant polynomial $P(x, y)$ such that $P(x + 2, y - 1) = P(x, y)$ for all x, y . Characterise all reachable positions using this invariant.

Very Hard – A Sequence Invariant

Given integer sequence $a_0, a_1, \dots$ with $a_0 = 0$, $a_1 = 1$, and $a_{n+1} = a_n + a_{n-1} + 1$ for $n \geq 1$, prove that

$$(-1)^{n-1}((a_n + 1)^2 - (a_{n-1} + 1)(a_{n+1} + 1)) = 1 \quad (n \geq 1).$$

7 Colouring And Tiling: Generalisations

Basic – Mutilated Chessboard: Four Corners Removed

An 8×8 chessboard has all four corner squares removed. Can the remaining board be tiled by dominoes? If so, give a tiling.

Easy – Tiling With T-Tetrominoes

Tile a 4×4 board by T-tetrominoes (four squares in a T shape). Extend your construction to every $4 \times (4k)$ board, $k \geq 1$.

Medium – Fault-Free Domino Tiling

A domino tiling of a rectangle has a fault line if an internal horizontal or vertical grid line from one side to the other cuts no dominoes. Prove that a 6×6 board has no fault-free tiling, but a 6×8 board does.

Hard – Triominoes On A Chessboard

Can an 8×8 chessboard with one corner removed be tiled by straight trominoes (1×3 rectangles)? Use a 3-colouring of the board.

Very Hard – Generalised Mutilated Chessboard

For which positive m, n can an $m \times n$ board *always* be tiled by dominoes after any two squares of opposite chessboard colours are removed? An empty tiling is allowed. Prove your characterisation.

8 Induction And Recursion: From Proof To Algorithm

Basic – Josephus Problem

n people stand in a circle. Eliminate every second person until one remains. Where should you stand to survive? Find a recurrence and closed form for $J(n)$.

Easy – Gray Codes

An n -bit Gray code lists all 2^n binary strings so consecutive strings differ in one bit. Give a recursive construction. Prove it works by induction.

Medium – Recursive 15-Puzzle Parity

Given a 4×4 15-puzzle configuration, design a recursive algorithm to determine if it is solvable. Use the invariant from Week 1.

Hard – Counterfeit Coin With n Weighings

Generalise the 12-coin, 3-weighing problem. With $n \geq 2$ weighings on a balance scale, what is the maximum number of coins among which you can find one counterfeit (heavy or light)? Give a strategy achieving the bound.

Very Hard – Hadamard Matrix Construction

A Hadamard matrix of order n is an $n \times n$ matrix with entries ± 1 whose rows are mutually orthogonal. Give a recursive construction for $n = 2^k$ and prove it works.

9 Algorithmic Bridges

Basic – A Maze As A Graph

You are given a rectangular grid of open and blocked cells, together with a start and finish cell. You may move one cell up, down, left, or right through open cells. Model the maze as a graph, describe how BFS finds the minimum number of moves, and explain why the answer is optimal.

Easy – Greedy Activity Selection

You have n half-open intervals $[s_i, f_i)$ with $s_i < f_i$. Choose a maximum-size subset of non-overlapping intervals. Prove that picking the interval with earliest finish time, then recursing, is optimal. (Exchange argument.)

Medium – DP From Induction: Coin Change

Given coin values $c_1, \dots, c_k$ and target n , find the minimum number of coins to make exactly n . Define $f(n)$ recursively, identify overlapping subproblems, and give a tabulation algorithm.

Hard – Sliding Puzzle State-Space Search

2×3 sliding puzzle with tiles 1–5 and one empty. From a given start state, find the minimum moves to reach the goal. Describe the state encoding, transitions, and why BFS works. (There are 360 reachable states from any fixed goal.)

Very Hard – Greedy Scheduling With Deadlines

n jobs, each takes 1 unit of time, has an integer deadline $d_i \geq 1$ and positive profit p_i . You may schedule at most one job per time slot. A job only earns profit if completed by its deadline. Design a greedy algorithm to maximise total profit and prove it correct. (Hint: consider jobs in decreasing profit order; use a disjoint-set union to find latest free slot.)

